

按键及开关检测

1、实验目的:

了解和检测开发板上的按键及开关特性。

2、实验原理:

图 1.1 开发板上的按键是按下去是接触为低电平，松开是断开为高电平。

图 1.2 中 SW1-SW4 是作为与 FPGA 的 IO 输入信号，拨上为“0”，拨下为“1”；

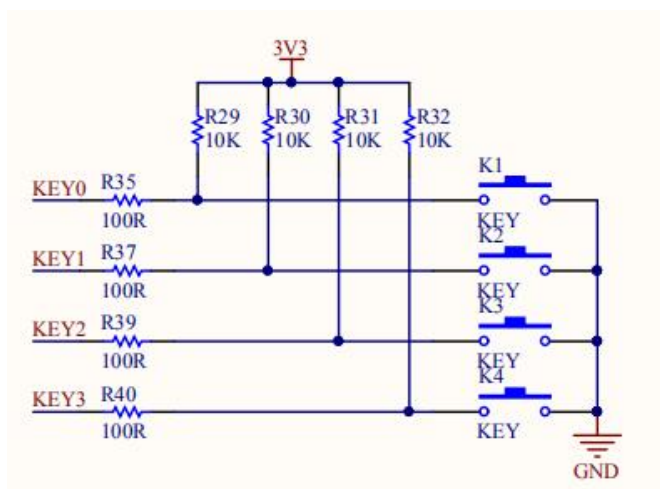


图 1.1

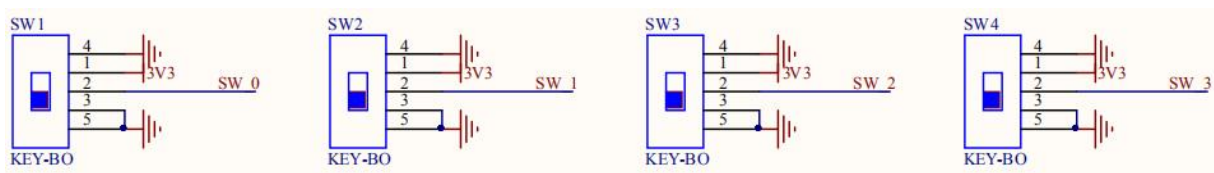


图 1.2

3、实验程序:

```

module key_test
(
  input clk, //时钟 50Mhz 板
  input [7:0] key, //输入 8 组信号
  output [7:0] led //LED 8 组
);
reg [7:0] led_r; //定义第一阶段寄存器，生成四个 D 触发器
reg [7:0] led_r1; //定义第二阶段寄存器，生成四个 D 触发器
  
```

```
always@(posedge clk)
begin
led_r <= ~key; // 第一级锁存数据
end
always@(posedge clk)
begin
led_r1 <= led_r; //第二级锁存数据
end
assign led = led_r1;
endmodule
```

5、引脚约束:

```
set_pin_assignment { clk } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[0] } { LOCATION = C8; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[1] } { LOCATION = C7; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[2] } { LOCATION = C6; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[3] } { LOCATION = C5; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[4] } { LOCATION = A2; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[5] } { LOCATION = B2; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[6] } { LOCATION = B1; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { key[7] } { LOCATION = C1; IOSTANDARD = LVCMOS33; PULLTYPE =
PULLUP; }
set_pin_assignment { led[0] } { LOCATION = C15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH
= 8; PULLTYPE = NONE; }
set_pin_assignment { led[1] } { LOCATION = C16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH
= 8; PULLTYPE = NONE; }
set_pin_assignment { led[2] } { LOCATION = B6; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
set_pin_assignment { led[3] } { LOCATION = A5; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =
8; PULLTYPE = NONE; }
```

```
set_pin_assignment { led[4] } { LOCATION = B5; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =  
8; PULLTYPE = NONE; }  
set_pin_assignment { led[5] } { LOCATION = A8; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =  
8; PULLTYPE = NONE; }  
set_pin_assignment { led[6] } { LOCATION = A7; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =  
8; PULLTYPE = NONE; }  
set_pin_assignment { led[7] } { LOCATION = A6; IOSTANDARD = LVCMOS33; DRIVESTRENGTH =  
8; PULLTYPE = NONE; }
```

6、下载测试：

分别按动四组按键与拨码开关，测试按键与LED灯的效果

